Website Image Generation Paths & Workflow Guide

Standard Operating Procedure: Image Storage Paths, Web URLs, Keyframe Chaining & Video Pipeline Integration

1. Overview: The Role of Images in AI Video Production

In modern generative video engineering (Google Flow, Veo 3.1 Fast/Quality, Gemini Omni Flash 1.1, and Seedance 2.5), generating high-fidelity still images is the foundational first step of the entire video production pipeline.

High-quality AI video requires decoupling style and motion into the Dual-Prompt Architecture:

• Prompt 1 (Keyframe Image Still): Governs the still visual identity, locked style string, subject facial features, wardrobe, setting, lighting setup, framing, and color palette.

• Prompt 2 (Video Motion & Audio): Governs only what moves: one camera move with a speed modifier, 1-3 sequenced action beats, the end state hold, and the audio acoustic line.

By generating Prompt 1 first and storing it at the designated website image path, the video generator takes this verified still image as its starting anchor, preventing identity drift, facial warping, and color inconsistency.

2. Master Architecture Table: Image Paths & Web URLs

All images are organized under the website's public static hierarchy (website\public\images\). Below is the complete architectural mapping for each image role:

Image Role / Purpose	Local Workspace Disk Path	Website Serving URL (HTTP)	Target Model / Tool
Prompt 1 Keyframes (First Frames)	website\public\images\keyframes\	http://localhost:3000/images/keyframes/<name>.png	Nano Banana Pro / Imagen 3 / Veo 3.1
Reference Ingredients (Characters/Products)	website\public\images\ingredients\	http://localhost:3000/images/ingredients/<name>.png	Google Flow Reference Ingestion
Chained Tail Frames (Shot 1 to Shot 2)	website\public\images\tails\	http://localhost:3000/images/tails/<name>.jpg	FFmpeg Frame Extractor (-sseof -0.1)
Timeline Thumbnails & UI Posters	website\public\images\thumbnails\	http://localhost:3000/images/thumbnails/<name>.png	Website Scenebuilder Timeline UI
Final Stitched Video Renders	website\public\renders\	http://localhost:3000/renders/<name>.mp4	FFmpeg Concat & Master Export

3. Step-by-Step Production Workflow for the Website

Step 1: Generate Keyframe Image Still (Prompt 1)

Using your image generation model (Nano Banana Pro, Nano Banana 2.1, or Imagen 3), execute Prompt 1 describing only the still scene attributes. Save the resulting image directly to:

Local File: website\public\images\keyframes\P01_S01_SH1_keyframe.png

```
HTTP URL:    http://localhost:3000/images/keyframes/P01_S01_SH1_keyframe.png
```

Step 2: Display Keyframe in the Website Scenebuilder Timeline

The Node.js website server (server.js) serves all files inside website\public as root static assets. In the website frontend (index.html), the image is displayed directly inside the timeline track using:

```
<div class="timeline-shot-card">`n  <img src="/images/keyframes/P01_S01_SH1_keyframe.png"
alt="Shot 1 Keyframe" />`n  <span>Shot 1 (4.0s)</span>`n</div>
```

Step 3: Feed Keyframe into Video Model as Start Frame (Image-to-Video)

• Google Veo 3.1 Fast / Quality: Set the mode to 'Frames to Video: First frame', passing P01_S01_SH1_keyframe.png as the reference, then execute Prompt 2 (Motion & camera dynamics).

• Gemini Omni Flash 1.1: Provide the keyframe as start_frame to lock actor facial structure, wardrobe colors, and scene lighting across the 10-second sequence.

Step 4: Extract Tail Frame for Chaining into Shot 2

To build a continuous 10.0-second scene (e.g., Shot 1 [4.0s] + Shot 2 [6.0s]) without visual jump, extract the exact last frame of Shot 1 using FFmpeg and save it into the tails directory:

```
ffmpeg -sseof -0.1 -i website\public\renders\P01_S01_SH1.mp4 -frames:v 1 -q:v 2
website\public\images\tails\P01_S01_SH1_tail.jpg
```

This extracted tail frame (served at http://localhost:3000/images/tails/P01_S01_SH1_tail.jpg) is then passed as the visual starting anchor for Shot 2 (6.0s), ensuring flawless character and lighting continuity across the internal cut.

4. Built-In Website API Endpoints for Images

The VIPA Studio website includes native REST API endpoints for image management:

• GET /api/images: Returns JSON containing disk paths, web URLs, and lists of all keyframes, ingredients, and thumbnails currently stored on the server.

• GET /api/status: Returns server health, VIPA Studio version 2.5, and active port.

• POST /api/split-prompt: Decomposes any raw text brief into Prompt 1 (Keyframe Image) and Prompt 2 (Video Motion).

5. Summary of Quick Paths for Developers

Local Workspace Root: D:\Open Code\Video production 261026 1st\self reserch with video

Keyframe Images Folder: website\public\images\keyframes

Ingredient Images Folder: website\public\images\ingredients

Tail Frames Folder: website\public\images\tails

Website URL: http://localhost:3000

```
Live Image API:           http://localhost:3000/api/images
```